UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE

Department of Computer Science

Software Engineering

Advanced Web Development

Hyper-scale Technical Management Limited - Team

Project Definition Document

Clinical Management and Session Control System

Group Members: Galarza César, Gualotuña Brayan, Erazo Isaac

NRC: 30716

Date: May 3, 2026

Project: Advanced Web System for "Ser salud" Comprehensive Clinic

Customer: Flores Villalva David Alejandro

"What is important is not being able, what is important is trying"

1. Problem Definition

The Comprehensive Rehabilitation Clinic "Ser salud" experiences critical operational bottlenecks due to an outdated manual methodology for appointment scheduling and session tracking. The administrative staff and specialists currently rely on informal communication channels, primarily WhatsApp, to coordinate time slots across multiple medical departments. This decentralized approach invariably leads to overlapping reservations, patient no-shows, and a significant loss of monetizable clinical hours. The absence of a unified digital ecosystem also complicates the tracking of patient cash flows and prepaid session packages. Manual ledgers fail to provide real-time data regarding session credits, introducing mathematical errors when calculating remaining balances. The proposed software solution must centralize these dispersed workflows into a cohesive digital platform, mitigating human error, optimizing the specialists' time, and establishing a rigorous financial control mechanism for the clinic's administrative department.

2. System Overview and Architecture

"Ser salud" operates under an interdisciplinary biopsychosocial approach, requiring a platform capable of handling diverse clinical specialties simultaneously. The proposed web system implements a robust, decoupled Client-Server architecture to centralize therapist calendars and patient data. This structural design ensures high performance and allows independent scaling of the user interface and the business logic layer. The system will communicate via a RESTful Application Programming Interface (API), enabling the frontend application to asynchronously fetch mock data during early development phases and seamlessly transition to a live database environment. The architecture supports secure environment configurations to protect sensitive routing and future third-party integrations.

Table 1: Customer Identification and Contact Data

Category	Details
Commercial Name	Ser salud Comprehensive Rehabilitation Clinic
Legal Representative	Flores Villalva David Alejandro
Tax ID (RUC)	1723008098001
Address	6 de Diciembre & Veintimilla
Contact Number	0979220237
Email	criflova@gmail.com

3. Comprehensive User Experience and Accessibility

The frontend architecture prioritizes immediate access to core scheduling functionalities to maximize patient conversion rates. The application initializes with a designated splash screen displaying the clinic's logo for three to five seconds, establishing a strong corporate identity before revealing the primary interface. The main landing page employs a zero-scroll philosophy, presenting all essential services and contact mechanisms within a single, responsive viewport. This layout explicitly targets mobile users who require rapid

interaction without navigating complex sub-menus. The system incorporates a dedicated emergency routing flow for urgent cases that cannot adhere to standard scheduling constraints. A floating action button redirects these specific users directly to a designated WhatsApp line, immediately displaying a digital disclaimer that outlines emergency rates and immediate payment conditions before the chat initiates.

4. Clinical Data Management and Security

Securing patient information and streamlining the intake process are critical components of the system's design. The initial registration phase collects minimal data to avoid user friction. The system auto-generates patient credentials using the first letter of their given name appended to their surname. The default password is set to the patient's national identification number (DNI), drastically reducing password recovery requests. The comprehensive biopsychosocial clinical history is decoupled from the initial account creation. Patients can complete this detailed form voluntarily within their private dashboard or synchronously with their specialist during the first consultation. This medical record tracks specific insurance provider affiliations, utilizing a controlled dropdown menu to identify entities like ISFA or Humana, which dictate distinct administrative reporting requirements. The system also features a specialized medication tracking module that strictly differentiates between permanent pharmacological treatments and situational prescriptions to prevent adverse reactions during physical therapies.

5. Clinical Workflow and Business Rules

Table 2: Medical Staff and Specialized Services

Specialist	**Areas of Expertise**
Flores Villalva David Alejandro	Traumatological, Sports, Neurological, Respiratory
Vasquez Camacho Juliana Lisseth	Traumatological, Pelvic Floor, Pre/Post-Prosthetic
Perez León Carolina Andrea	Traumatological, Respiratory Rehabilitation

5.1 Scheduling and Cross-Department Logic

The booking engine enforces strict chronological validations to maintain the integrity of the clinical agenda. Appointments require formal scheduling or cancellation with a minimum advance notice of 24 hours. The system actively blocks overlapping reservations across the entire facility, ensuring a patient cannot hold two simultaneous appointments in different departments. Specialists possess full modification rights over their designated calendars and maintain read-only access to other departments. This cross-department visibility enables therapists to coordinate internal patient referrals directly during a session without relying on the administrative reception.

5.2 Transaction and Payment Processing

The current payment flow mandates direct bank transfers to minimize operational costs associated with external payment gateways. Patients receive the clinic's official banking

details immediately upon reserving a time slot in the calendar. The system registers the appointment under a "Pending" status. Administrative staff or the assigned specialist must manually verify the transfer receipt via the established contact channels. Once verified, the authorized personnel update the appointment status to "Confirmed" within the system dashboard.

6. Scalability and Future Integrations

The decoupled nature of the application lays the groundwork for modular expansions without requiring fundamental system overhauls. The current manual payment verification process is designed as a placeholder for automated financial workflows. The system architecture supports the future integration of direct banking APIs to automate transaction reconciliations instantly. The administrative module is built to dynamically accept new medical specialties and register incoming clinical staff as the business scales. This flexibility ensures the software adapts to the Customer's operational growth without incurring excessive redevelopment costs.

7. System Metrics and Calculations

The system automates the calculation of session credits for patients purchasing extended treatment packages. The **Remaining Balance (RB)** algorithm updates dynamically after each completed visit:

$$RB = TP - (SC \times UC)$$

Where:

- **TP:** Total Amount Paid for the package upfront.
- **SC:** Number of Sessions Completed and marked as attended by the specialist.
- **UC:** Unit Cost predetermined per individual therapy session.

Performance monitoring relies on aggregating data to evaluate operational efficiency:

1. **Clinic Occupation Rate:** The exact percentage of available clinical time slots effectively reserved and executed by the staff.
2. **Financial Liability Balance:** The aggregated sum of all financial advances paid by active patients corresponding to therapy sessions that are still pending execution.
3. **Penalty Tracking:** Automated logging of canceled appointments or no-shows occurring within the restricted 24-hour window to enforce non-refundable deposit policies.

This advanced web platform ensures that "Ser salud" optimizes its administrative overhead while providing a seamless, secure, and professional digital experience for both its patients and medical specialists.

UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE

Department of Computer Science

Software Engineering

Advanced Web Development

Hyper-scale Technical Management Limited - Team

User Stories Document

Features & Acceptance Criteria Breakdown

Group Members:
- Galarza César (Scrum Master)
- Gualotuña Brayan (Lead Developer)
- Erazo Isaac (Software Architect / UI-UX)

NRC: 30716

Date: May 3, 2026

Project: Advanced Web System for "Ser salud" Comprehensive Clinic

Customer: Flores Villalva David Alejandro

"What is important is not being able, what is important is trying"

1. Introduction

This document details the breakdown of system features into actionable User Stories for the Ser salud Clinical Management System. Following the Scrum framework, the project scope is divided into 10 main Epics (Features), tailored specifically to the operational, clinical, and administrative requirements defined by the Customer during the requirement elicitation phase. These features are further decomposed into specific User Stories containing precise Acceptance Criteria, Estimations, and System Behaviors.

2. Feature 1: Authentication & Role Access

This feature manages system access, ensuring that patients and clinic staff have appropriate permissions within the platform while avoiding complex password recovery loops.

2.1 US-01: Simplified Patient Registration

ID	US-01
Role	As a New Patient, I want to create an account using my ID (DNI) as my password, so I do not forget my login credentials in the future.
Estimation (Points)	5
Priority	High (1)
Acceptance Criteria	
Context	Public Landing Page / Login Portal.
Event	• User inputs primary data: Full Name and DNI. • System validates DNI uniqueness. • System auto-generates username (First initial + Last name, e.g., dflores) and sets DNI as the permanent password.
Specific Behavior	The system creates the profile with minimal friction and immediately redirects the user to the Patient Dashboard to begin scheduling.
Adverse Behavior	If the DNI is already registered, a validation error prompts the user to log in instead of creating a duplicate record.
Responsible	Galarza César (Scrum Master)

2.2 US-02: Administrator Access

ID	US-02
Role	As the Clinic Administrator, I want to assign specific roles to my staff so that therapists only manage their designated departments.
Estimation (Points)	3
Priority	High (1)
Acceptance Criteria	
Context	Admin Control Panel.
Event	• Admin navigates to the 'User Management' module. • Selects an existing user and elevates their privileges to 'Specialist'. • Assigns a specific therapy area (e.g., Traumatology) to the new specialist.
Specific Behavior	The selected user gains read/write access to the specific department's calendar and read-only access to the rest of the clinic's schedule.
Adverse Behavior	The system rejects the role assignment if the specialist is not mapped to an existing active therapy category.
Responsible	Brayan Gualotuña (Lead Developer)

3. Feature 2: Dynamic UI & Accessibility

Focuses on delivering a highly responsive, zero-scroll interface that caters to mobile users and emergency situations.

3.1 US-03: Splash Screen & Zero-Scroll Landing

ID	US-03
Role	As a Public User, I want to see a clear, concise landing page with a loading splash screen, so I can understand the clinic's services instantly without scrolling.
Estimation (Points)	5
Priority	Medium (2)
Acceptance Criteria	
Context	Web Application Initialization.
Event	• User navigates to the URL. • System displays a 3-5 second Splash Screen featuring the 'Ser salud' logo. • Transitions smoothly to a consolidated, zero-scroll main menu.
Specific Behavior	All available therapies, specialist directory, and the 'Book Appointment' button are fully visible in a single viewport, optimized for mobile devices.
Adverse Behavior	If the browser's viewport is extremely small, the CSS framework automatically stacks the elements to maintain readability without horizontal scrolling.
Responsible	Isaac Erazo (Software Architect / UI-UX)

3.2 US-04: Emergency Routing System

ID	US-04
Role	As a Patient with an urgent condition, I want to contact a specialist directly for a same-day appointment bypassing the 24-hour restriction.
Estimation (Points)	3
Priority	High (1)
Acceptance Criteria	
Context	Main Dashboard - Floating Action Button.
Event	• User clicks the 'Emergency Appointment' floating button. • System triggers a modal window detailing the emergency protocols and extra fees. • User accepts the terms and is redirected to WhatsApp.
Specific Behavior	The system bypasses the standard booking engine and establishes a direct line with the administrative triage number.
Adverse Behavior	If the user clicks 'Cancel' on the disclaimer modal, the redirection is aborted, and they remain on the standard scheduling page.
Responsible	Brayan Gualotuña (Lead Developer)

4. Feature 3: Clinical Intake & Medical History

Replaces physical paperwork with a deferred, highly detailed digital intake form capable of tracking specific medications and insurance parameters.

4.1 US-05: Deferred Clinical History

ID	US-05
Role	As a Patient, I want to fill out my detailed medical history later or with my therapist, so I am not overwhelmed during my initial registration.
Estimation (Points)	8
Priority	High (1)
Acceptance Criteria	
Context	Patient Dashboard - 'My Medical History' Module.
Event	• Patient logs in and sees a non-blocking prompt to complete their medical history. • Patient clicks 'Skip' or 'Fill Now'.
Specific Behavior	If skipped, the prompt will reappear gently before the next appointment. The specialist can also access this exact form from their terminal during the physical session.
Adverse Behavior	The system allows saving partial progress in the form so the patient does not lose data if they exit the module prematurely.
Responsible	Galarza César (Scrum Master)

4.2 US-06: Pharmacological Tracking

ID	US-06
Role	As a Therapist, I need to know if a patient is taking permanent or situational medications, so I can adapt the physical therapy intensity safely.
Estimation (Points)	5
Priority	Medium (2)
Acceptance Criteria	
Context	Clinical History Form - Medication Section.
Event	• User indicates they are currently medicated. • System displays radio buttons: 'Permanent (e.g., Hypertension)' or 'Situational (e.g., Muscle Relaxant)'. • User details the medication.
Specific Behavior	The data is prominently flagged in the Therapist's daily agenda view, ensuring the specialist is warned before the session begins.
Adverse Behavior	If the user selects 'Yes' to medication but leaves the text field blank, the form highlights the error and prevents submission.
Responsible	Isaac Erazo (Software Architect / UI-UX)

5. Feature 4: Advanced Agenda & Cross-Visibility

Provides the clinic staff with tools to manage their working hours and view the broader operational flow of the facility.

5.1 US-07: Specialist Cross-Department View

ID	US-07
Role	As a Specialist, I want to view the calendars of other departments so I can coordinate treatments without administrative bottlenecks.
Estimation (Points)	5
Priority	Medium (2)
Acceptance Criteria	
Context	Specialist Dashboard.
Event	• Therapist selects a 'View Clinic Agenda' tab. • System fetches all scheduled appointments across all departments for the day.
Specific Behavior	The therapist can see exactly which time slots are occupied in Occupational Therapy or Neurology but cannot modify or delete those external appointments (Read-Only access).
Adverse Behavior	Unauthorized attempts to edit a cross-department appointment result in a strict HTTP 403 Forbidden alert.
Responsible	Isaac Erazo (Software Architect / UI-UX)

5.2 US-08: Internal Patient Referral

ID	US-08
Role	As a Therapist, I want to book an appointment for my current patient in another department directly from my terminal.
Estimation (Points)	8
Priority	High (1)
Acceptance Criteria	
Context	Active Session Interface.
Event	• Therapist clicks 'Refer Patient'. • Selects 'Neurology' and picks an available slot for the following week. • Confirms booking on behalf of the patient.
Specific Behavior	The system generates the appointment, linking it to the patient's existing ID, and automatically notifies the Neurology department of the internal referral.
Adverse Behavior	The therapist cannot book the referral if the target department's time slot violates the 24-hour advance notice rule.
Responsible	Brayan Gualotuña (Lead Developer)

6. Feature 5: Booking Engine & Chronological Constraints

The central algorithm that prevents agenda collisions and enforces the clinic's preparation timeframes.

6.1 US-09: Strict 24-Hour Booking Restriction

ID	US-09
Role	As the System, I need to prevent bookings less than 24 hours in advance to provide specialists with adequate preparation time.
Estimation (Points)	5
Priority	High (1)
Acceptance Criteria	
Context	System Time Validation Engine.
Event	• Patient selects a desired date on the interactive calendar. • System calculates the delta between the server's current timestamp and the requested slot.
Specific Behavior	If the time difference is less than 24 hours, the slot is visually grayed out and rendered unclickable by the frontend framework.
Adverse Behavior	Any attempt to bypass the frontend restriction via direct API calls is blocked by the backend validation logic.
Responsible	Galarza César (Scrum Master)

6.2 US-10: Cross-Department Overlap Prevention

ID	US-10
Role	As the System, I must prevent a patient from booking two different therapies at the exact same hour to avoid physical conflicts in the clinic.
Estimation (Points)	8
Priority	High (1)
Acceptance Criteria	
Context	Booking Submission Logic.
Event	• Patient holds an active booking for 'Traumatology' at 10:00 AM on Tuesday. • Patient attempts to book 'Respiratory Therapy' for 10:00 AM on the same Tuesday.
Specific Behavior	The system queries the patient's DNI against all departments, detects the collision, and displays an error: 'You already have an active session scheduled for this exact time.'
Adverse Behavior	The system suggests the nearest available adjacent slot (e.g., 11:00 AM) to the user.
Responsible	Brayan Gualotuña (Lead Developer)

7. Feature 6: Appointment Modification & Penalties

Allows patient autonomy while strictly safeguarding the clinic's revenue against last-minute cancellations.

7.1 US-11: 24-Hour Cancellation Policy Enforcement

ID	US-11
Role	As the System, I must enforce the cancellation policy so the clinic does not lose revenue on no-shows or late cancellations.
Estimation (Points)	5
Priority	High (1)
Acceptance Criteria	
Context	Appointment Cancellation Flow.
Event	• User clicks 'Cancel Appointment' in their dashboard. • System checks if the appointment is within the next 24 hours.
Specific Behavior	If $<$ 24 hours, a mandatory modal appears: '100% penalty applies. Your advanced transfer will not be refunded. Do you wish to proceed?'.
Adverse Behavior	If the user accepts, the slot is freed, but the financial ledger marks the previous deposit as 'Earned Revenue / Penalty' rather than a liability.
Responsible	Brayan Gualotuña (Lead Developer)

7.2 US-12: Mandatory Cancellation Reason

ID	US-12
Role	As an Administrator, I want patients to provide a reason for cancellation so I can track operational issues or patient dissatisfaction.
Estimation (Points)	3
Priority	Low (3)
Acceptance Criteria	
Context	Appointment Cancellation Flow.
Event	• User proceeds with a valid cancellation ($>$ 24 hours). • A text area appears requiring justification (e.g., 'Work conflict', 'Health issue').
Specific Behavior	The cancellation cannot be finalized until the text area contains at least 10 characters.
Adverse Behavior	The provided reason is logged in the database and visible in the Admin's analytics dashboard.
Responsible	Isaac Erazo (Software Architect / UI-UX)

8. Feature 7: Manual Financial Processing

Handles the specific requirement of processing payments via direct bank transfers, avoiding credit card gateways.

8.1 US-13: Bank Transfer Registration (Pending State)

ID	US-13
Role	As a Patient, I want to receive the clinic's bank details and upload my transfer receipt so I can secure my booking.
Estimation (Points)	8
Priority	High (1)
Acceptance Criteria	
Context	Checkout & Confirmation Phase.
Event	• User finalizes slot selection. • System displays 'Ser salud' Bank Account numbers and the exact 50% advance amount required. • User uploads a screenshot/PDF of the transfer.
Specific Behavior	The system registers the appointment in the database with a strict 'Pending Verification' status, temporarily holding the slot.
Adverse Behavior	If no receipt is uploaded within a 2-hour window, the system's cron job automatically frees the slot back to the public pool.
Responsible	Galarza César (Scrum Master)

8.2 US-14: Manual Payment Verification (Confirmed State)

ID	US-14
Role	As an Admin/Specialist, I want to review uploaded transfer receipts to manually confirm the appointments.
Estimation (Points)	5
Priority	High (1)
Acceptance Criteria	
Context	Admin Control Panel - Pending Payments.
Event	• Admin logs in and opens the 'Pending Verifications' queue. • Admin clicks on a patient's uploaded receipt to view the image. • Admin clicks 'Verify & Confirm'.
Specific Behavior	The appointment status instantly changes from 'Pending' to 'Confirmed', triggering a confirmation notification to the patient.
Adverse Behavior	If the receipt is invalid or fake, the Admin clicks 'Reject', which instantly frees the slot and notifies the patient of the failed transaction.
Responsible	Brayan Gualotuña (Lead Developer)

9. Feature 8: Session Packages & Credits

Manages the logic for patients who purchase rehabilitation packages (e.g., 10 sessions) for long-term treatments.

9.1 US-15: Deduct Completed Session from Package

ID	US-15
Role	As a Therapist, I want to mark a session as completed so the system deducts it from the patient's prepaid package.
Estimation (Points)	3
Priority	Medium (2)
Acceptance Criteria	
Context	Therapist Daily Agenda.
Event	• Therapist finalizes a physical session. • Therapist clicks 'Mark as Completed' on the specific appointment card.
Specific Behavior	The system updates the Patient's overall ledger, decrementing the 'Remaining Sessions' counter by exactly 1 unit.
Adverse Behavior	The system prevents the deduction if the patient's package balance is already at 0, prompting a 'Requires New Purchase' alert.
Responsible	Brayan Gualotuña (Lead Developer)

9.2 US-16: Display Remaining Balance

ID	US-16
Role	As a Patient, I want to see my available session credits clearly so I know exactly when I need to purchase a new package.
Estimation (Points)	2
Priority	Low (3)
Acceptance Criteria	
Context	Patient Dashboard.
Event	• User logs into the dashboard. • System queries the remaining sessions logic: $RB = TP - (SC \times UC)$.
Specific Behavior	Displays a prominent numeric counter: 'You have X sessions remaining in your current plan'.
Adverse Behavior	If the calculated balance reaches 0, the dashboard renders a direct call-to-action button to 'Purchase New Package'.
Responsible	Isaac Erazo (Software Architect / UI-UX)

10. Feature 9: Automated Communications & Vouchers

Replaces informal messaging with professional, automated system notifications and verifiable digital receipts.

10.1 US-17: Downloadable PDF Appointment Voucher

ID	US-17
Role	As a Patient, I want to download a PDF voucher of my confirmed booking to present at the clinic's reception.
Estimation (Points)	5
Priority	Medium (2)
Acceptance Criteria	
Context	Patient Dashboard - Appointment Details.
Event	• User clicks 'Download Digital Receipt' on a 'Confirmed' appointment. • The frontend utilizes a library to render a PDF document.
Specific Behavior	The PDF includes the 'Ser salud' logo, Specialist Name, Date, Time, Transfer Amount Verified, and the Patient's DNI.
Adverse Behavior	If the browser blocks the PDF download, a digital QR code containing the appointment hash is generated on-screen as a fallback.
Responsible	Isaac Erazo (Software Architect / UI-UX)

10.2 US-18: Insurance-Specific Documentation Export

ID	US-18
Role	As a Specialist, I want to export a specific clinical report format depending on the patient's insurance (e.g., ISFA vs Humana).
Estimation (Points)	8
Priority	Medium (2)
Acceptance Criteria	
Context	Specialist Dashboard - Patient Record.
Event	• Therapist finalizes session notes. • Clicks 'Export Report'. • System checks the patient's registered insurance provider.
Specific Behavior	If 'ISFA' is detected, the system generates a monthly aggregated report format. If 'Humana' is detected, it generates a single-session signature format.
Adverse Behavior	If the patient has no insurance, the button defaults to a standard, generic clinic discharge summary.
Responsible	Brayan Gualotuña (Lead Developer)

11. Feature 10: Administrative Configuration

Provides the clinic owner (Customer) with the tools to scale the business by adding new departments and monitoring metrics.

11.1 US-19: Dynamic Creation of Therapies

ID	US-19
Role	As an Administrator, I want to create new therapy categories (e.g., Nutrition, Psychology) so the system scales as the clinic grows.
Estimation (Points)	5
Priority	High (1)
Acceptance Criteria	
Context	Admin Control Panel - Services.
Event	• Admin clicks 'Add New Service'. • Enters 'Psychology', sets a default session duration (e.g., 60 mins), and assigns an active specialist.
Specific Behavior	The new service immediately populates the public frontend dropdown menu, allowing patients to begin booking sessions without requiring developer intervention.
Adverse Behavior	The system prevents the creation of a new service if a responsible specialist is not linked during the creation process.
Responsible	Galarza César (Scrum Master)

11.2 US-20: Financial Liability Tracking

ID	US-20
Role	As an Admin, I want to track the Total Financial Liability Balance to know exactly how much money we owe in pending prepaid sessions.
Estimation (Points)	5
Priority	High (1)
Acceptance Criteria	
Context	Admin Control Panel - Analytics.
Event	• Admin views the Financial Metrics tab. • System aggregates all advance payments associated with 'Pending' or upcoming 'Confirmed' sessions.
Specific Behavior	Displays the total monetary value of sessions owed to patients, distinctly separated from 'Earned Revenue' (completed sessions).
Adverse Behavior	Cancelled appointments that triggered the 24-hour penalty are automatically moved from 'Liabilities' to the 'Earned Revenue' column.
Responsible	Isaac Erazo (Software Architect / UI-UX)

UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE

Department of Computer Science

Software Engineering

Advanced Web Development

Hyper-scale Technical Management Limited - Team

Product Backlog Document

Comprehensive Requirements & PBI Breakdown

Group Members: Galarza César, Gualotuña Brayan, Erazo Isaac

NRC: 30716

Date: May 3, 2026

Project: Advanced Web System for "Ser salud" Comprehensive Clinic

Customer: Flores Villalva David Alejandro

"What is important is not being able, what is important is trying"

1. Introduction and Product Vision

The following Product Backlog serves as the primary artifact for the requirements engineering phase of the "Ser salud" Clinical Management System. Driven by Scrum methodologies, this document consolidates all functional, visual, and business requirements elicited directly from the Customer into structured Product Backlog Items (PBIs). The vision is to deliver a humane, highly accessible, and technically robust web platform that eliminates manual administrative bottlenecks while preserving a warm and straightforward experience for patients seeking physical rehabilitation.

2. Product Backlog Items (PBIs) by Epic

2.1 Epic 1: Fluid User Experience & Emergency Routing

Description: Ensures the system is visually welcoming, mobile-responsive without excessive scrolling, and provides immediate solutions for patients in critical condition.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
1.1	Splash Screen & UI	**Desc:** Implement a 3-5 second loading splash screen displaying the clinic's logo (Tesla/Nike style) before rendering a zero-scroll main menu. **AC:** Must fit entirely on a standard mobile screen without vertical scrolling.	5	High
1.2	Emergency Contact Button	**Desc:** A floating WhatsApp button for same-day urgent appointments bypassing the 24-hour rule. **AC:** Must trigger a disclaimer modal explaining emergency fees and conditions before redirecting the user to WhatsApp.	3	High
1.3	Dynamic Services Catalog	**Desc:** Display the clinical services (Traumatological, Neurological, Respiratory, Sports) directly on the landing page for quick access.	2	Med

2.2 Epic 2: Streamlined Authentication & Roles

Description: Focuses on reducing patient friction during registration and defining clear boundaries between administrative and specialist access.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
2.1	Frictionless Patient Login	**Desc:** Registration asks only for DNI and Full Name to prevent user drop-off. **AC:** System auto-generates a username (e.g., initial + surname) and sets the DNI as the permanent password.	5	High
2.2	Specialist Profiles	**Desc:** Creation of therapist accounts linked to specific departments. **AC:** Therapists have read/write access to their agenda and read-only cross-visibility to other departments.	5	High
2.3	Admin Privilege Management	**Desc:** Administrator dashboard to manage clinic staff and services. **AC:** Admin can add new therapy areas and assign specialists dynamically.	3	Med

2.3 Epic 3: Deferred Clinical Intake & Insurance

Description: Decouples the dense medical history from the initial login, allowing patients to provide crucial health context at their own pace or alongside the specialist.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
3.1	Deferred Medical History	**Desc:** A non-mandatory clinical history form presented as a gentle prompt before the first booking. **AC:** Can be skipped by the patient and filled out later by the therapist during the physical session.	8	High
3.2	Pharmacological Tracking	**Desc:** Specific fields within the clinical history to track active medications. **AC:** Must explicitly differentiate between "Permanent" (e.g., hypertension) and "Situational" (e.g., muscle relaxants) medications.	5	High
3.3	Insurance Mapping	**Desc:** A dropdown list mapping the patient to specific insurance providers. **AC:** Includes options like ISFA or Humana to dictate if the specialist needs to generate specific monthly reports or single-session signatures.	5	Med

2.4 Epic 4: Strict Booking Engine & Referrals

Description: The core scheduling algorithm that respects the clinic's preparation times and manages internal traffic to prevent physical overlapping.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
4.1	24-Hour Chronological Validation	**Desc:** Enforce a strict buffer for all non-emergency appointments. **AC:** The interactive calendar disables any time slots that are less than 24 hours away from the server's current timestamp.	8	High
4.2	Cross-Department Anti-Overlap	**Desc:** Prevents a single patient from holding concurrent appointments. **AC:** If Patient A has Traumatology at 10:00 AM, the system blocks them from booking Neurology at 10:00 AM on the same day.	8	High
4.3	Internal Patient Referral	**Desc:** Allows a specialist to schedule a subsequent appointment for their patient. **AC:** Therapists can book slots in their own calendar or refer the patient directly into an available slot in another department.	5	Med

2.5 Epic 5: Manual Transactions & Cancellations

Description: Secures the clinic's revenue by managing manual bank transfers, deposit verifications, and enforcing strict cancellation penalties.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
5.1	Bank Transfer Verification Flow	**Desc:** Appointments are created as "Pending" until a 50% advance is verified. **AC:** Displays clinic bank details. Patient uploads receipt. Admin/Specialist manually approves the receipt to change status to "Confirmed".	8	High
5.2	Cancellation Penalty Enforcement	**Desc:** Logic to penalize late cancellations. **AC:** Canceling within 24 hours triggers a non-refundable 100% penalty warning. The deposit is logged as earned revenue.	5	High
5.3	Mandatory Cancellation Reason	**Desc:** Require justification when a patient cancels a confirmed slot. **AC:** Patient must input at least 10 characters explaining the cancellation reason before the system frees the time block.	3	Low

2.6 Epic 6: Session Packages & Digital Vouchers

Description: Automates the mathematical tracking of prepaid therapies and generates professional outputs for patients.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
6.1	Automated Session Deduction	**Desc:** Tracks the remaining balance of prepaid 10-session packages. **AC:** When a therapist marks a session "Completed", the patient's remaining sessions counter decrements by 1. Reaching 0 prompts a renewal alert.	5	High
6.2	Downloadable PDF Vouchers	**Desc:** Generate a digital receipt upon appointment confirmation. **AC:** PDF must include the Ser salud logo, Date, Time, Amount, Specialist, and a reminder to arrive 15 minutes early.	5	Med
6.3	Document Uploads	**Desc:** Allow patients to attach previous external lab results. **AC:** If digital files are unavailable, the UI must persistently remind the patient to bring physical copies to the first session.	3	Low

2.7 Epic 7: Administrative Configuration & System Scalability

Description: Provides the Customer (Clinic Owner) and administrative staff with total autonomy to scale the business without requiring further developer intervention, allowing the dynamic addition of new medical branches and personnel.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
7.1	Dynamic Service Creation	**Desc:** Allows the Admin to create new therapy categories (e.g., Psychology, Nutrition) dynamically to expand the clinic's offerings. **AC:** The new service immediately populates the public frontend dropdown menu. The system strictly prevents service creation if a responsible specialist is not assigned.	5	High
7.2	Staff Roster Management	**Desc:** Interface for adding, suspending, or updating specialist profiles and departments. **AC:** Suspended specialists are immediately hidden from the public booking engine, but their historical session data remains intact in the database for auditing.	5	Med
7.3	Global Variables Configuration	**Desc:** Admin module to update the clinic's global variables, such as the WhatsApp emergency number or official Bank Account. **AC:** Changes reflect instantly across all patient-facing interfaces and payment checkout screens without requiring a server reboot.	3	Low

2.8 Epic 8: Financial Analytics & Operational Metrics

Description: Transforms raw scheduling data into actionable business intelligence, allowing the Customer to monitor the clinic's financial health, liability in prepaid sessions, and overall staff efficiency.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
8.1	Financial Liability Dashboard	**Desc:** A visual analytics tab calculating the clinic's exact financial debt regarding prepaid but unfulfilled therapy sessions. **AC:** Aggregates all 50% advance payments of 'Pending' and 'Confirmed' upcoming sessions into a distinct 'Liabilities' visual chart.	8	High
8.2	Penalty Revenue Tracking	**Desc:** Automated financial tracking of deposits retained due to the 24-hour cancellation rule. **AC:** When a late cancellation occurs, the system automatically subtracts the deposit from the 'Liability' column and adds it to an 'Earned Penalty Revenue' ledger.	5	High
8.3	Clinic Occupation Rate	**Desc:** Evaluates the efficiency of the specialists by comparing booked hours versus available working hours. **AC:** Displays a dynamic Pie Chart to the Admin, filtering out manually blocked hours (like lunch breaks) so they do not negatively skew the efficiency percentage.	5	Med

2.9 Epic 9: Clinical Document Management

Description: Focuses on the secure handling of external medical files and the internal logging of post-therapy observations to build a comprehensive, cross-departmental patient record.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
9.1	External Exam Uploads	**Desc:** Allows patients to securely attach laboratory results, X-Rays, or doctor referrals (PDF/JPG) during the intake phase. **AC:** Files over 5MB are rejected with a clear UI error. Uploaded files are directly linked to the specific appointment ID.	8	Med
9.2	Physical Document Reminder	**Desc:** A persistent UI nudge for patients who bypass the digital document upload phase. **AC:** If no file is attached, the confirmation screen and the PDF Voucher must explicitly print: "Please bring your physical medical documents to your first session."	3	Low
9.3	Post-Session Observation Ledger	**Desc:** A dedicated text area for therapists to log treatment details, patient progress, and specific physical reactions after a session. **AC:** This ledger is permanently attached to the patient's ID and is fully visible to other specialists if the patient is internally referred to another department.	5	High

2.10 Epic 10: Patient Engagement & Automated Flow

Description: Implements gamification strategies and automated reminders to optimize the clinic's time, reducing the administrative burden on the staff.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
10.1	Profile Completion Incentive	**Desc:** A gamified prompt to encourage patients to fill out the dense medical history from home, saving 30 minutes of clinical time. **AC:** The dashboard displays a banner offering a predefined discount or benefit on their next package if they complete their profile to 100% digitally.	5	Low
10.2	Pre-Appointment Reminders	**Desc:** System automation to ensure patients do not forget their scheduled and confirmed therapies. **AC:** A cron job triggers an automated alert 12 hours prior to the session, explicitly reminding the patient to arrive 15 minutes early to optimize clinical flow.	5	Med
10.3	Post-Therapy Satisfaction Nudge	**Desc:** A brief feedback mechanism to ensure quality of service and patient retention. **AC:** 24 hours after a completed session, the system prompts the user in their dashboard to rate their experience out of 5 stars, flagging any 1-star reviews for the Admin.	3	Low

UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE

Department of Computer Science

Software Engineering

Advanced Web Development

Hyper-scale Technical Management Limited - Team

Official Meeting Minutes

First Customer Review & Requirements Elicitation

Group Members: Galarza César, Gualotuña Brayan, Erazo Isaac

NRC: 30716

Date of Meeting: May 3, 2026

Project: Advanced Web System for "Ser salud" Comprehensive Clinic

Customer: Flores Villalva David Alejandro

"What is important is not being able, what is important is trying"

1. Meeting Metadata

Attribute	Details
Meeting Type	Initial Prototype Review and Requirements Elicitation
Duration	1 Hour 25 Minutes (Recorded Session)
Customer Representative	Flores Villalva David Alejandro (Physiotherapist & Clinic Owner)
Scrum Team Attendees	Galarza César (Scrum Master), Gualotuña Brayan (Lead Developer), Erazo Isaac (Software Architect / UI-UX)
Objective	To present the initial frontend landing page and booking logic mockup, gather direct feedback from the Customer, refine business rules, and establish concrete technical parameters for the next development sprints.

2. Executive Summary

The meeting focused on validating the initial architectural approach and user interface design for the "Ser salud" Clinical Management System. The Customer provided extensive feedback regarding user accessibility, demanding a highly simplified, zero-scroll landing interface that immediately directs patients to the booking engine. Significant discussions were held regarding clinical business rules, resulting in the establishment of a strict 24-hour advance booking policy, the decoupling of the comprehensive medical history from the initial registration phase, and the implementation of a manual verification flow for bank transfers. Furthermore, the Customer outlined the necessity for a multi-specialist environment where therapists can view cross-departmental agendas and perform internal patient referrals. The development team agreed to incorporate these specifications into the Product Backlog for immediate execution.

3. Detailed Discussion & Customer Feedback

3.1 User Interface & Frontend Accessibility

The Customer explicitly requested that the public-facing landing page be as minimalistic as possible to avoid cognitive overload for incoming patients. He proposed the implementation of a brief "Splash Screen" (lasting 3 to 5 seconds) displaying the clinic's logo prior to loading the main interface, citing modern corporate websites like Tesla or Nike as references. Furthermore, the Customer emphasized a "zero-scroll" philosophy, asking the development team to condense all vital information—such as the services catalog, professional directory, and the appointment booking button—into a single viewport. To facilitate immediate action, the team proposed a floating action button that remains visible at all times, redirecting users either to the booking engine or a direct WhatsApp line for emergencies. Visually, the Customer requested replacing the initial placeholder

background with a transparent, representative image and updating the branding icons to match the clinic's identity.

3.2 User Authentication & Registration Flow

A critical pain point identified by the Customer was the likelihood of patients forgetting complex passwords, leading to an influx of account recovery requests. To mitigate this, the Customer and the development team agreed upon a simplified authentication mechanism. The username will be automatically generated using a combination of the patient's first initial and last name, while the default password will strictly be the patient's national identification number (DNI). Additionally, the registration form will be heavily abbreviated to ask only for essential demographic and contact data. The comprehensive medical history, which includes dense clinical questions, will be moved to a secondary, optional phase to prevent user drop-off during the initial sign-up process.

3.3 Clinical Intake & Medical History Management

The management of the patient's medical history was extensively debated. The Customer noted that conducting an initial clinical interview physically consumes up to 30 minutes of a therapy session. Therefore, having the patient fill this out digitally beforehand is highly valuable. However, making it a mandatory step during registration causes friction. The agreed solution is a "Deferred Medical History" module. The system will prompt the user to complete their history from their dashboard optionally, and the clinic may offer gamified incentives (e.g., a service discount) to encourage completion.

Specific medical data points were also defined. The system must capture the patient's exact insurance provider via a dropdown list (including options like ISFA or Humana), as different insurances require drastically different reporting formats from the therapists. Additionally, the medication tracking section must force the patient to differentiate between "Permanent" medications (such as blood pressure regulators) and "Situational" drugs (such as temporary muscle relaxants or painkillers), as this directly impacts the physical therapy methodology applied during the session.

3.4 Scheduling Constraints & Emergency Protocols

The Customer established strict temporal boundaries for the booking engine. A standard appointment must be scheduled with a minimum of 24 hours of advance notice, automatically blocking the current and immediately adjacent days in the system calendar. To address the reality of acute, sudden injuries, the team will implement an "Emergency Appointment" pathway. This flow bypasses the 24-hour rule by redirecting the patient directly to the clinic's administrative WhatsApp. Before redirection, the system will trigger a mandatory disclaimer modal explicitly stating the conditions, increased rates, and immediate payment requirements for same-day emergency slots.

Furthermore, the system must include robust anti-overlap logic. If a patient books a session in Traumatology at a specific hour, the system must globally block that patient's DNI from booking any other therapy (e.g., Occupational Therapy) at that exact same time to prevent physical scheduling conflicts within the clinic.

3.5 Payment Processing & Cancellation Penalties

To avoid external transaction fees from credit card gateways, the Customer mandated that all payments be processed via direct bank transfers. The system will display the clinic's bank account details upon the finalization of the booking flow. Appointments will remain in a "Pending" state within the database until the patient sends the transfer receipt to the administrative WhatsApp line, at which point a staff member or the specialist will manually update the status to "Confirmed".

Regarding cancellations, the Customer requires a stringent penalty system to protect the clinic's revenue. If a patient attempts to cancel a confirmed appointment with less than 24 hours of notice, the system will display an unbypassable warning stating that the advanced deposit is 100% non-refundable. Additionally, all cancellations will require the patient to input a written justification in a text area, providing the administration with actionable analytics regarding patient drop-off reasons.

3.6 Specialist Roles & Internal Referrals

The system must support a multi-specialist architecture, as the clinic plans to scale its services rapidly. Each therapist will possess a dedicated login granting full write access to their specific department's calendar. Crucially, the Customer requested a "cross-visibility" feature: specialists must have read-only access to the agendas of all other departments. This allows therapists to seamlessly coordinate internal referrals. For example, a Traumatologist finishing a session can actively view the Neurology calendar and directly book a follow-up appointment for their patient in that department without needing to route the patient back through the main administrative reception.

4. Action Items & Next Steps

#	Action Item Description	Assigned To
1	Update the global Product Backlog to reflect all elicited requirements, specifically the 24-hour rule, manual payment flow, and deferred clinical history.	Galarza César (Scrum Master)
2	Refactor the frontend architecture to support the 3-5 second Splash Screen and the zero-scroll layout for the main landing page.	Erazo Isaac (UI/UX)
3	Implement the emergency WhatsApp redirection flow, including the mandatory disclaimer modal for same-day appointments.	Gualotuña Brayan (Lead Developer)
4	Restructure the database schemas to support the differentiation between "Permanent" and "Situational" medications in the clinical history.	Gualotuña Brayan (Lead Developer)
5	Provide the development team with the definitive list of accepted Medical Insurances (e.g., ISFA, Humana) to populate the frontend dropdowns.	Flores David (Customer)

5. Specialized Clinical Workflows and Exceptions

During the requirements elicitation phase, the Customer highlighted several operational edge cases that diverge from the standard "happy path" of the booking engine. To ensure the software is highly resilient and adaptable to real-world clinical environments, the system must account for these specific out-of-band operations and data persistence scenarios.

5.1 In-Clinic Assisted Onboarding (Digital Divide Segments)

The Customer emphasized that certain demographics, particularly elderly patients or those lacking digital literacy, may be unable to navigate the web-based scheduling module or complete the deferred clinical history independently. To prevent a regression to physical paper files and ensure the database remains fully synchronized, the system incorporates an "Assisted Onboarding Flow." In this scenario, the clinic's administrative staff or the attending specialist can bypass the public portal and manually instantiate the patient's profile from an elevated-privilege terminal. The staff will directly input the patient's demographic data, auto-generate the DNI-based credentials, and transcribe the clinical history in real-time during the physical visit. This guarantees that all patient data remains centralized in the cloud infrastructure regardless of the user's technological proficiency.

5.2 Specialist-Driven Appointment Negotiations

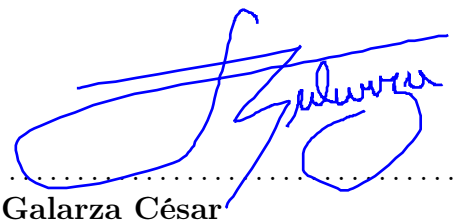
While the standard scheduling flow places unverified appointments into a "Pending" state, the Customer specified a critical operational exception: a specialist might identify a logistical conflict or determine a more optimal time block for a specific therapy after the patient has requested a slot. Instead of executing a hard rejection, the system interface provides the specialist with the capability to propose an "Alternative Time Slot." This action triggers a negotiation loop where the patient is notified of the modified schedule and can either accept the new terms or cancel the request entirely. This flexibility empowers the medical staff to dynamically optimize their daily agendas without losing the patient's booking intent.

5.3 Interdisciplinary Medical Ledger and Contraindications

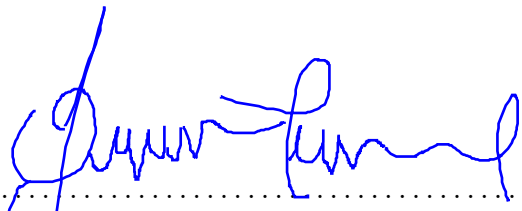
Because "Ser salud" operates under an interdisciplinary model, patients frequently receive concurrent treatments across different departments. The Customer noted severe clinical risks associated with isolated information silos—for instance, subjecting a patient to strenuous traumatological therapy immediately after they have ingested a strong muscle relaxant prescribed by another specialist. To strictly prevent such medical contraindications, the system features a permanent "Post-Session Observation Ledger." Upon completing any therapy session, the attending specialist is mandated to digitally log the specific procedures applied, the patient's physical reactions, and any temporary medications ingested. This ledger is persistently attached to the patient's unified profile and is instantly visible to any other specialist consulting the file, establishing a safe and cohesive continuum of care across the entire facility.

Edge Case Scenario	System Resolution / Workflow	Responsible Actor
Lack of Digital Literacy	Patient cannot use the web app. Profile and clinical history are created manually via the internal administrative dashboard.	Receptionist or Specialist
Sub-optimal Slot Request	The requested time is unviable. The system allows the therapist to hold the request and propose an alternative time for patient approval.	Assigned Specialist
Missing Transfer Receipt	A 2-hour window expires without payment verification. The system's cron job automatically frees the time block back to the public pool.	Automated System
Ignored Medical History	The patient skips the digital intake prompt. The profile is flagged, and the system alerts the therapist to collect the data physically.	Assigned Specialist
Contraindicated Medication	Patient declares usage of strong situational drugs (e.g., somniferous relaxants). The system flags the profile to adjust physical therapy intensity.	Assigned Specialist

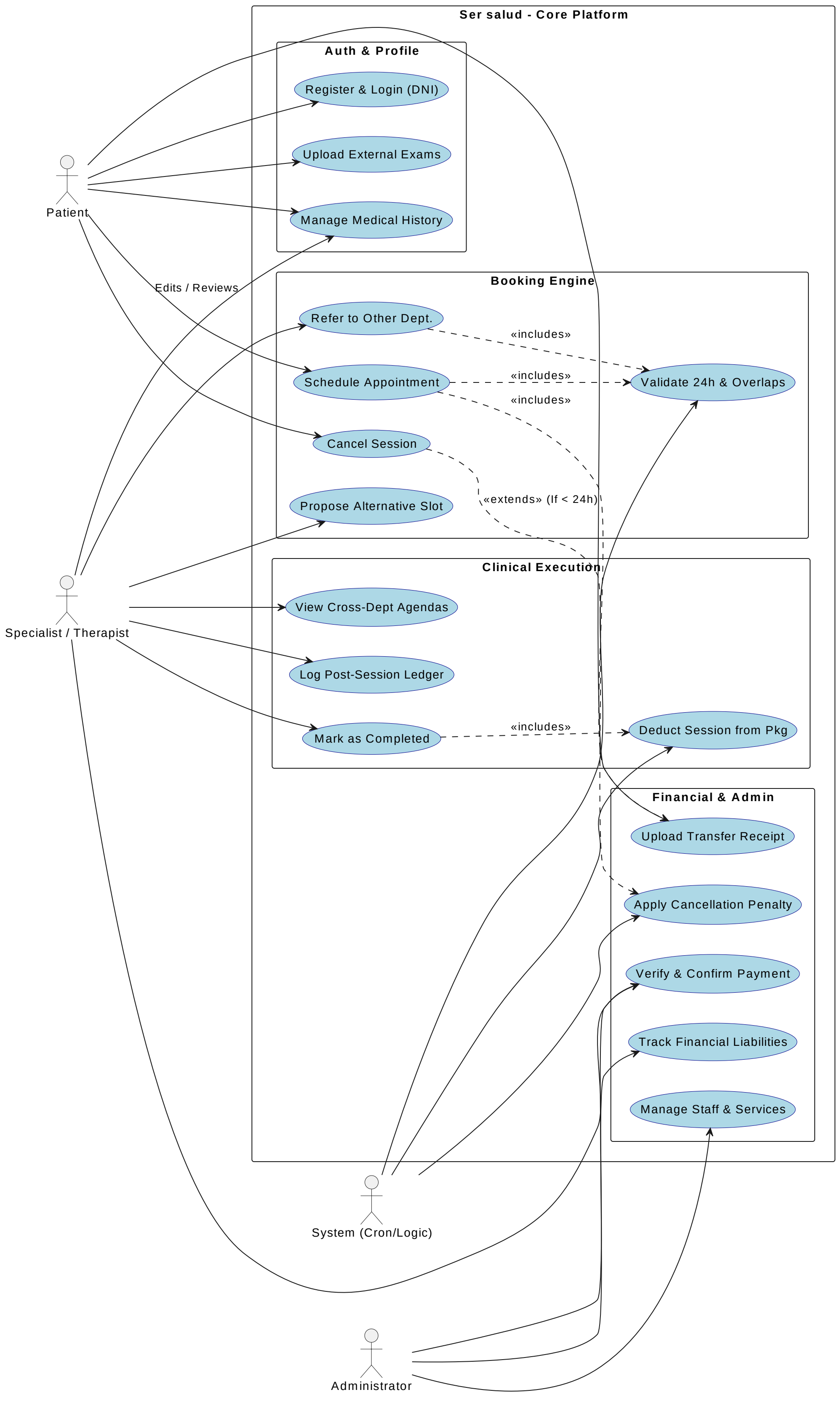
Signatures of Approval:

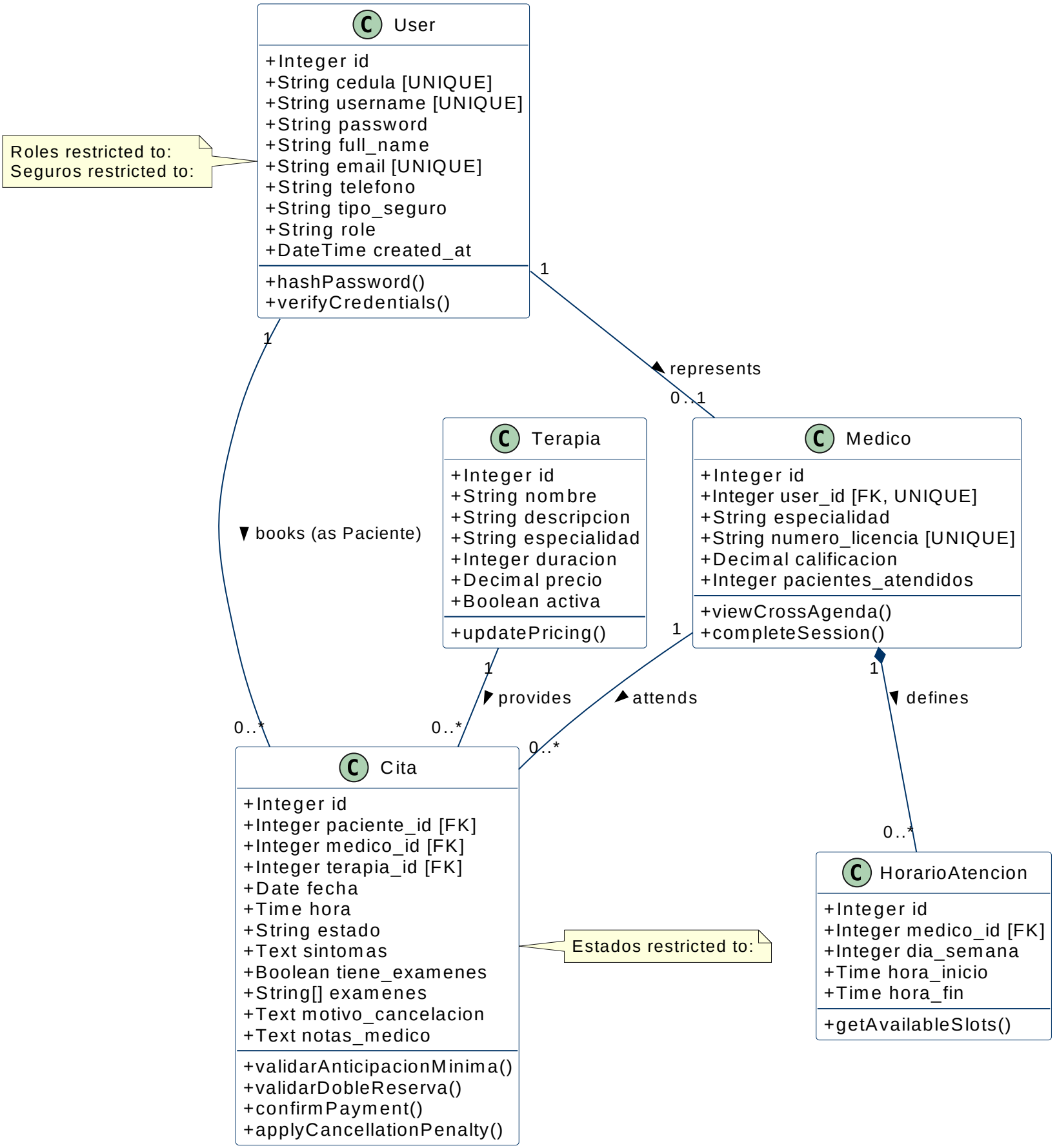


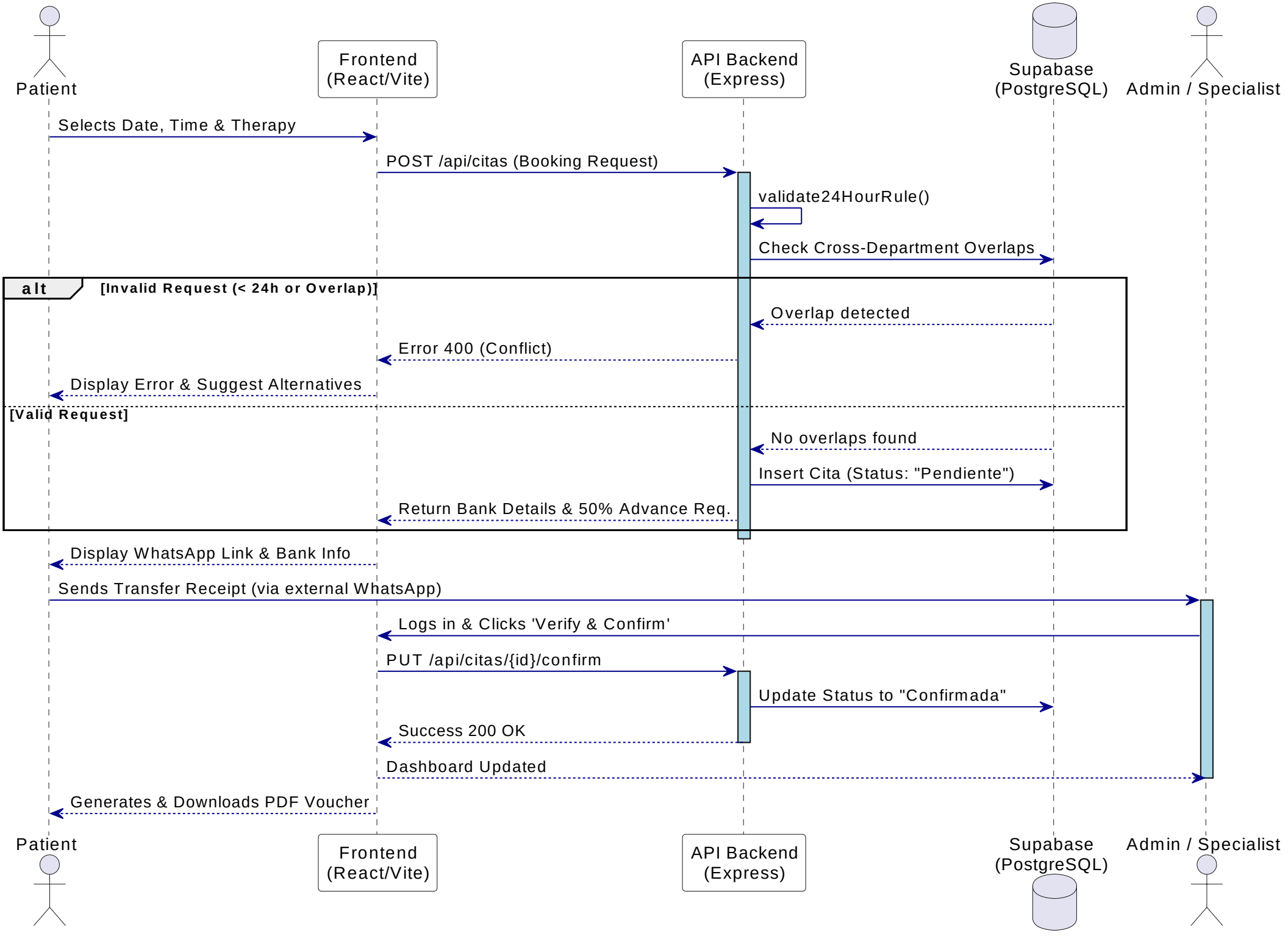
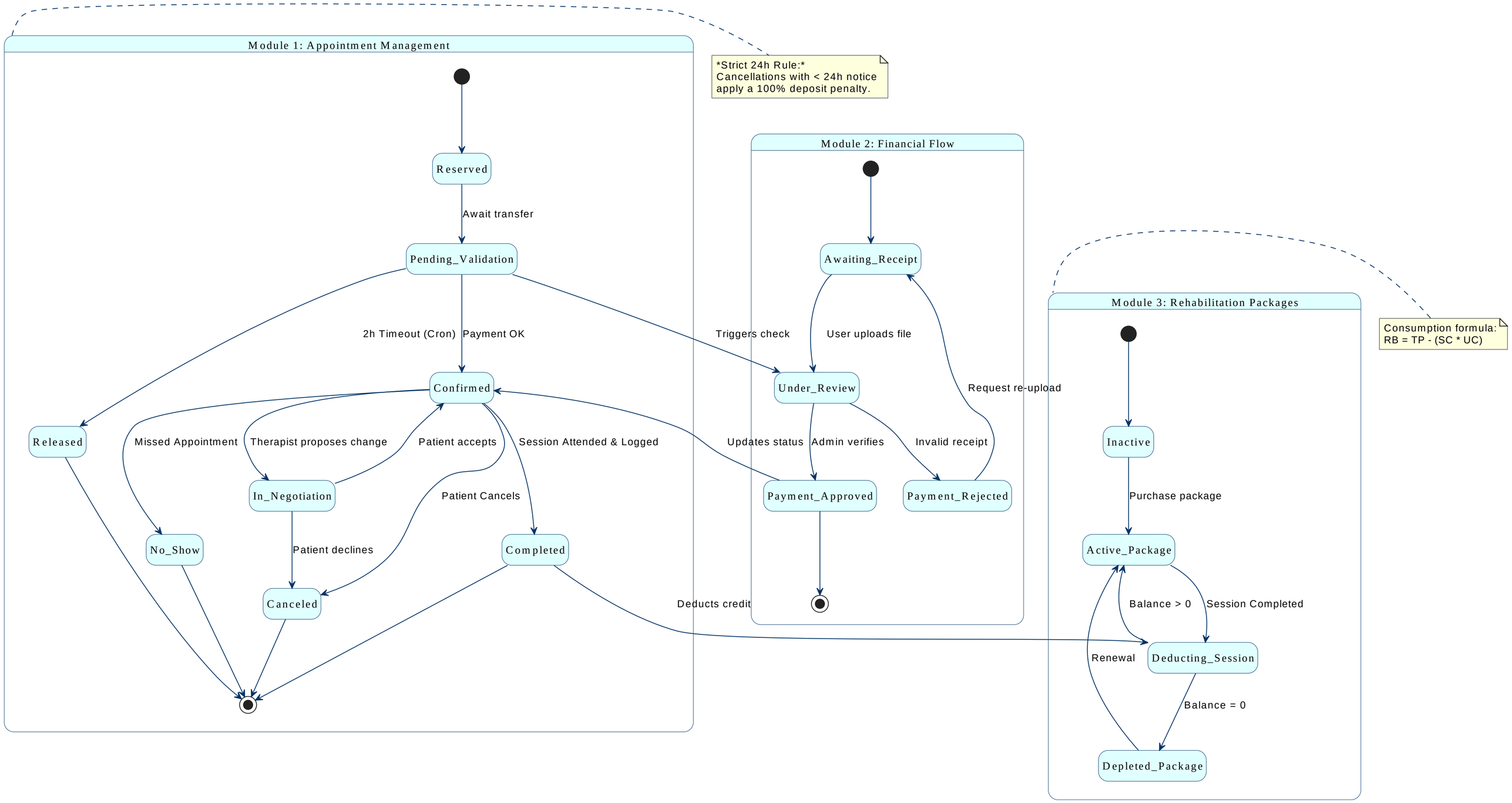
Galarza César
Scrum Master / HTML Team



Flores Villalva David Alejandro
Clinic Owner / Customer


Ser salud - Core Platform
Auth & Profile
Register & Login (DNI)
Upload External Exams
Manage Medical History
Patient
Edits / Reviews
Booking Engine
Refer to Other Dept.
«includes»
Schedule Appointment
«includes»
Validate 24h & Overlaps
«includes»
Cancel Session
«extends» (If < 24h)
Propose Alternative Slot
Clinical Execution
View Cross-Dept Agendas
Specialist / Therapist
Log Post-Session Ledger
Mark as Completed
«includes»
Deduct Session from Pkg
Financial & Admin
Upload Transfer Receipt
Apply Cancellation Penalty
Verify & Confirm Payment
Track Financial Liabilities
Manage Staff & Services
System (Cron/Logic)
Administrator


User
+Integer id
+String cedula [UNIQUE]
+String username [UNIQUE]
+String password
+String full_name
+String email [UNIQUE]
+String telefono
+String tipo_seguro
+String role
+DateTime created_at
+hashPassword()
+verifyCredentials()
Roles restricted to:
Seguros restricted to:
1
represents
0..1
1
books (as Paciente)
0..*
Terapia
+Integer id
+String nombre
+String descripcion
+String especialidad
+Integer duracion
+Decimal precio
+Boolean activa
+updatePricing()
Medico
+Integer id
+Integer user_id [FK, UNIQUE]
+String especialidad
+String numero_licencia [UNIQUE]
+Decimal calificacion
+Integer pacientes_atendidos
+viewCrossAgenda()
+completeSession()
1
provides
0..*
1
attends
0..*
1
defines
0..*
Cita
+Integer id
+Integer paciente_id [FK]
+Integer medico_id [FK]
+Integer terapia_id [FK]
+Date fecha
+Time hora
+String estado
+Text sintomas
+Boolean tiene_examenes
+String[] examenes
+Text motivo_cancelacion
+Text notas_medico
+validarAnticipacionMinima()
+validarDobleReserva()
+confirmPayment()
+applyCancellationPenalty()
Estados restricted to:
HorarioAtencion
+Integer id
+Integer medico_id [FK]
+Integer dia_semana
+Time hora_inicio
+Time hora_fin
+getAvailableSlots()


Sequence Diagram: Booking & Manual Payment Flow
Patient
Frontend (React/Vite)
API Backend (Express)
Supabase (PostgreSQL)
Admin / Specialist
Selects Date, Time & Therapy
POST /api/citas (Booking Request)
validate24HourRule()
Check Cross-Department Overlaps
alt
[Invalid Request (< 24h or Overlap)]
Overlap detected
Error 400 (Conflict)
Display Error & Suggest Alternatives
[Valid Request]
No overlaps found
Insert Cita (Status: "Pendiente")
Return Bank Details & 50% Advance Req.
Display WhatsApp Link & Bank Info
Sends Transfer Receipt (via external WhatsApp)
Logs in & Clicks 'Verify & Confirm'
PUT /api/citas/{id}/confirm
Update Status to "Confirmada"
Success 200 OK
Dashboard Updated
Generates & Downloads PDF Voucher
Patient
Frontend (React/Vite)
API Backend (Express)
Supabase (PostgreSQL)
Admin / Specialist

Integrated State Machine - Ser salud